Research Statement

CIFRE PhD — Generali France / Investment Division

"When Diversification Fails"

Regime-Conditional Factor Dependence, Supervised Capital Proxy, and Deep Hedging for Insurance Total Portfolio Approach

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1. INDUSTRIAL CONTEXT AND MOTIVATION

1.1. Starting point : 2022, a dependence crisis

The year 2022 was a major stress test for insurance asset managers. Unlike previous crises — which were essentially isolated shocks on a single asset class — 2022 saw equities and bonds fall simultaneously, inter-asset correlations turn positive, and traditional asset-class diversification break down entirely. For a life insurer such as Generali France, this episode exposed a fundamental structural flaw : Strategic Asset Allocation (SAA) models assume stable, linear dependence structures between economic factors, whereas the empirical evidence shows that these dependencies reverse abruptly in stress regimes.

This observation directly motivates the object of this thesis : *how to manage an insurance balance sheet when factor dependence becomes unstable, non-linear, and regime-dependent ?*

1.2. The current SAA process and its limitations

How the SAA process works today. The current SAA process at Generali France operates in three sequential steps, each carrying structural limitations.

Step 1 — Economic Scenario Generator (ESG). A proprietary ESG produces thousands of economic trajectories (interest rates, credit spreads, equity, real estate, inflation) over a long horizon, calibrated on historical data with **fixed correlations averaged across all regimes**. These trajectories feed all downstream actuarial calculations.

Step 2 — SCR calculation via nested LSMC. The Solvency Capital Requirement (SCR) is the regulatory capital under Solvency II, defined as the 99.5% one-year Value-at-Risk of net asset value change. Its calculation requires evaluating the Best Estimate of Liabilities (BEL) — the risk-neutral present value of all future policyholder cash flows — under each of the 10,000+ real-world scenarios produced by the ESG. Since each BEL evaluation requires a nested risk-neutral simulation (typically 1,000 inner paths), the raw computational cost is $10{,}000 \times 1{,}000 = 10$ million simulations. The industry-standard solution — Least-Squares Monte Carlo (LSMC, Krah et al., 2018) — replaces the inner simulation by a polynomial proxy $\hat{V}^L(\mathbf{x}) = \sum_k \beta_k \phi_k(\mathbf{x})$ fitted offline on a design-of-experiment. Even so, running Prophet / MoSes for a full SCR calculation takes **several hours**, making dynamic optimisation loops computationally infeasible.

Step 3 — Annual Markowitz optimisation and manual execution. The SCR is treated as a fixed constraint in a mean-variance optimisation over asset classes, revised **once per year**. Rebalancing instructions are transmitted **manually by email or Excel spreadsheet to GenAM** (Generali Asset Management, the delegated manager), which then executes trades over several days. In addition, the optimisation criterion is fixed *ex ante* : there is no automatic identification of the dominant portfolio objective — for example duration control, inflation protection, carry harvesting, or capital efficiency — even though the relevant risk factor varies across portfolios and across regimes.

Four structural limitations.

1. **Regime-blind dependence.** The ESG calibrates correlations on long historical averages. In 2022, the equity/bond correlation switched from -0.3 (diversifying) to $+0.6$ (concentration risk) within weeks. A model with fixed correlations cannot anticipate such reversals : the Markowitz optimum for a low-inflation regime is a high-risk portfolio in a stagflation regime.
2. **LSMC proxy limitations.** The polynomial LSMC proxy is accurate only within the region covered by the design-of-experiment, which is constructed without regime conditioning. Precision degrades precisely in the stress zones most relevant for capital decisions. Moreover, the proxy is not expressed in terms of TPA factors, preventing its direct integration into a factor-based allocation loop.
3. **Static, siloed decision-making.** Markowitz optimises a single-period mean-variance criterion without anticipating future regime transitions or internalising hedging costs. Allocation and hedging are decided separately, yielding a suboptimal joint policy for an insurer whose critical losses are convex. Moreover, the objective function is imposed uniformly rather than identified automatically from the portfolio's dominant risk factor and balance-sheet context, which is precisely what a dynamic TPA process should adapt over time.
4. **Private asset valuation.** Private equity, private debt, infrastructure and real estate constitute a growing share of Generali's portfolio. Their quarterly, smoothed NAV valuations create an **illusion of low correlation** with public markets : in 2008, private equity NAVs remained stable for 6–12 months after the crash before correcting sharply. This artificial smoothing *understates* true economic dependence in stress regimes and introduces a lag bias in the SCR calculation. Furthermore, the irrevocable capital commitments of closed-end funds are incompatible with dynamic rebalancing frameworks that treat all assets as freely tradeable.

1.3. The TPA as target framework : what is missing

The **Total Portfolio Approach (TPA)** is the management framework that Generali France aims to adopt to progressively replace the current asset-class SAA at local level. Rather than optimising over asset classes, TPA evaluates each decision through its marginal contribution to total balance sheet risk via **common economic factors** : Duration, Inflation, Growth, Liquidity, Carry. Empirical studies on sovereign wealth funds that have already adopted TPA (Future Fund, CPP Investments) report approximately +1% excess return per year relative to classical SAA (Thinking Ahead Institute, 2025 ; CAIA, 2024).

However, TPA currently rests on simplistic dependence assumptions : linear, Gaussian, non-regime-conditional. This thesis provides the missing quantitative foundations to make TPA operational for an insurer under Solvency II. The **regime-conditional factor dependence** is the transversal thread : it is first modelled and simulated (Axis 1), then translated into fast capital costs via a supervised proxy (Axis 2), and finally integrated into a joint allocation-hedging policy (Axis 3). The concrete deliverable : a **Python library** deployed in production at Generali.

Extension to liability-side and joint factors. The TPA framework as currently applied — including in the sovereign wealth fund implementations serving as reference (Future Fund, CPP Investments) — operates exclusively on **asset-side factors** (Duration, Inflation, Growth, Liquidity, Carry). For an insurance company, this is structurally insufficient : the balance sheet is jointly driven by factors that exclusively affect liabilities (surrender rate dynamics, longevity, policyholder behaviour) and by **cross-cutting factors** that simultaneously affect both the asset and liability sides. The most prominent example is the **yield curve** : a parallel shift simultaneously reprices the bond portfolio *and* the Best Estimate of Liabilities, creating complex convexity interactions that cannot be captured by an asset-only factor decomposition. This thesis therefore extends the TPA factor space to include these joint factors, providing a genuinely *balance-sheet-wide* factor framework.

A what-if toolbox for allocation decisions. A key operational consequence of modelling factor trajectories stochastically (Axis 1) is the ability to explore the full spectrum of extreme scenarios systematically — adverse shocks (stress regimes : stagflation, credit crunch, equity crash) as well as scenarios more favourable than expected (yield normalisation, above-trend growth). This produces an **allocation reaction function** : a structured mapping from any projected factor scenario — whether originating from the asset side, the liability side, or joint cross-cutting factors such as the yield curve — to the optimal portfolio response. The result is a *what-if plan* computed ex ante, enabling the investment team to anticipate all market configurations rather than reacting to them ex post. This reaction function is a direct output of the Axis 3 joint policy optimisation and constitutes one of the principal industrial deliverables of the thesis.

A central scenario for governance and committee use. Alongside the full distribution of generated scenarios, the thesis also aims to extract a **central scenario** — for example a regime-conditional medoid or barycentric representative path — that serves as the baseline case for committee discussions, budgeting, and comparison with adverse and favourable scenarios. The goal is not to replace the stochastic distribution, but to provide a single auditable reference scenario around which allocation decisions and deviations can be explained.

2. RESEARCH QUESTIONS

Research Question 1 (Regime-conditional factor dependence : from GAN to formal guarantees)**.** *How to simulate the joint distribution of TPA factors conditionally on the current macroeconomic regime — identified in real time by a sequential filter — while capturing the non-linear, asymmetric tail dependencies relevant for extreme risk ? Specifically : (i) how to formally validate the conditional dependence structure learned by a regime-conditional GAN using copula specification tests ; and (ii) how to obtain W_2-convergence guarantees when upgrading to a Score-Based Generative Model (SGM) with sequentially estimated regime states ?*

Research Question 2 (Supervised SCR proxy for dynamic decision-making)**.** *How to construct a proxy for the insurance SCR — expressed as a function of TPA factors, the current regime, and key liability states including private asset NAV lags — that is sufficiently accurate in decision-relevant zones (near the regulatory threshold, under stress regimes), with controlled error bounds per regime and sub-10ms inference time, enabling its integration into a dynamic optimisation loop as an explicit capital constraint in the allocation programme ?*

Research Question 3 (Joint allocation and deep hedging policy)**.** *How to learn a dynamic decision policy jointly optimising TPA factor exposures and explicit hedging positions (deep hedging), with a portfolio-specific objective function automatically inferred from the dominant risk factor to optimise, and under capital and liquidity constraints including an explicit SCR proxy constraint and private asset engagement limits, in the presence of unstable, regime-conditional factor dependencies, so as to reduce the tail risk of the insurance balance sheet ?*

3. RELATED WORK AND POSITIONING

3.1. Generative models and regime-conditional dependence

Generative Adversarial Networks applied to financial series have demonstrated their capacity to capture multivariate stylized facts : heavy tails, volatility clustering, and asymmetries (Wiese et al., 2020, Quant-GAN ; Yoon et al., 2019, TimeGAN). Conditional GAN architectures (cGAN) allow generation to be conditioned on external variables — here, the macroeconomic regime. However, GANs suffer from *mode collapse* and provide no formal guarantees on the quality of the generated distribution, limiting their regulatory auditability.

Copula theory (Sklar, 1959) provides a complementary framework for validating learned dependence. By Sklar's theorem :

$$P(\mathbf{F}_t \leq \mathbf{f}) = C(F_1(f_1), \ldots, F_d(f_d)),$$

where C is the copula capturing the dependence structure independently of the marginals. Fermanian (2013, 2022) has developed specification tests for conditional copulas, providing the statistical tools to formally validate that a GAN has correctly captured the conditional tail dependence coefficient λ_k per regime.

Score-Based Generative Models (SGMs) offer a generative alternative with formal convergence guarantees. Strasman, Ocello et al. (TMLR, 2024) establish explicit bounds on the Wasserstein-2 distance as a function of the noise schedule $g(t)$:

$$\mathrm{KL}(\hat{p} \,\|\, p^*) \;\leq\; \int_0^T g(t)^2 \, \varepsilon_{\text{score}}(t) \, dt + \varepsilon_{\text{init}}. \tag{1}$$

Gentiloni-Silveri and Ocello (ICML, 2025) extend these guarantees to non-log-concave distributions — notably Gaussian mixtures corresponding to distinct market regimes.

Identified gap. Neither conditional GANs nor conditional SGMs have been applied to simulating the joint distribution of TPA factors with real-time regime identification. For the GAN : the absence of formal conditional tail dependence validation. For the SGM : the absence of W_2 bounds for the conditional case when the regime state is estimated sequentially with inference error. Both contributions constitute Axis 1.

3.2. Insurance liability proxy models

The LSMC framework of Krah et al. (2018) is the industry reference for life insurance proxy models, approximating the BEL by polynomial regression on a pre-computed design-of-experiment. This approach presents two critical limitations : it does not condition the SCR on the current macroeconomic regime, and its accuracy degrades precisely in the stress zones most relevant for the allocation decision. Neural network approaches (Huber & Schmocker, 2022) have demonstrated that a well-specified proxy can achieve 1–3% accuracy with inference times below 10ms.

Identified gap. Existing proxies are constructed independently of the regime and TPA factors, and none accounts for the NAV-lag bias of private assets in the SCR calculation. A regime-conditioned, factorially coherent proxy with adaptive sample weighting and controlled error bounds does not exist in the literature. This is the contribution of RQ2.

3.3. Deep RL, dynamic allocation and deep hedging

Wekwete (2023) demonstrated the applicability of Deep RL (PPO, SAC) to insurance ALM in stochastic environments. Buehler et al. (2019) introduced *deep hedging* : learning an optimal hedging strategy via a neural network minimising a convex risk measure, without market completeness assumptions. This approach has been extended to extreme risk hedging (Carbonneau, 2021 ; Cao et al., 2023) and multi-asset portfolios under stress regimes.

Identified gap. Learning a joint allocation-and-hedging policy — fed by an unstable dependence model and a fast SCR proxy, respecting private asset engagement constraints, while automatically selecting the relevant portfolio objective in factor space and being validated through out-of-sample allocation backtests within a TPA insurance framework under Solvency II — does not exist in the literature. This is the contribution of RQ3.

4. METHODOLOGY AND DELIVERABLES

4.1. Axis 1 — Regime-conditional factor dependence (Years 1–2)

Axis 1 proceeds in two methodological stages corresponding to a progressive increase in theoretical rigour.

Stage 1 — Regime-conditional GAN (Year 1). Let $z_t \in \{1, \ldots, K\}$ be the macroeconomic regime identified in real time by a particle filter (SMC, Chopin & Papaspiliopoulos, 2020). A conditional GAN is trained to simulate the joint distribution of TPA factors conditionally on the current regime :

$$\mathbf{F}_{t+1} \mid z_t = k \;\sim\; G_\theta(\varepsilon, k), \quad \varepsilon \sim \mathcal{N}(0, I), \tag{2}$$

where G_θ is the generator and the discriminator D_ϕ distinguishes real from synthetic scenarios regime by regime. The advantage of the GAN is total flexibility : no parametric assumption on the dependence structure is imposed, allowing the model to learn asymmetries and non-linearities in tail dependence under stress. The conditional tail dependence coefficient

$$\lambda_k = \lim_{u \to 1} P\big(F_i > F_{i,k}^{-1}(u) \mid F_j > F_{j,k}^{-1}(u),\, z_t = k\big)$$

is estimated empirically and validated via Fermanian's (2013) specification tests for conditional copulas.

Private asset treatment. The factor vector includes a *vintage lag factor* $\tilde{F}_t^{\mathrm{PA}} = \mathrm{NAV}_{t-1}^{\mathrm{PA}}$ capturing the quarterly smoothing bias of private equity, real estate and infrastructure NAVs. The GAN learns the true economic dependence between private and public assets, including the delayed correction dynamic observed in 2008 and 2020.

Stage 2 — SGM with W_2 guarantees (Year 2). The GAN, while empirically powerful, suffers from mode collapse and provides no formal guarantees — limiting regulatory auditability. The second stage replaces or complements the GAN with a conditional SGM, building on the theoretical framework of Strasman, Ocello et al. (TMLR, 2024) and Gentiloni-Silveri & Ocello (ICML, 2025). The conditional OU forward process for regime $z_t = k$ is :

$$d\mathbf{x}^\tau = -\mathbf{x}^\tau\, d\tau + \sqrt{2}\, g_k(\tau)\, dW_\tau, \quad \tau \in [0, T], \tag{3}$$

with regime-specific noise schedule $g_k(\tau)$. The convergence bounds give :

$$W_2(\hat{p}_k,\, p_k^*)^2 \;\leq\; C \int_0^T g_k(\tau)^2\, \varepsilon_{\mathrm{score}}(\tau)\, d\tau,$$

even for non-log-concave distributions corresponding to distinct market regimes.

Theoretical contribution. (i) Empirical validation of tail dependence λ_k generated by the GAN via Fermanian (2013) specification tests. (ii) Extension of the W_2 bounds of Gentiloni-Silveri & Ocello (ICML 2025) to the conditional SGM case with online HMM — the key open question : how do these bounds evolve when regime z_t is estimated sequentially with particle filter inference error ?

Scenario selection layer. Beyond generating a full scenario cloud, Axis 1 also includes a scenario selection step producing a **central regime-conditional scenario** used as the baseline path for reporting, comparison, and downstream optimisation diagnostics.

Python deliverable — `factor_engine` module.

- `HMMRegimeFilter` : online SMC filter, sequential update of $P(z_t \mid \mathbf{F}_{1:t})$ on real-time Bloomberg data.
- `RegimeConditionedGAN` : conditional GAN trained per regime ; tail dependence validation via copula specification tests (Fermanian, 2013).
- `RegimeConditionedSGM` : conditional SGM with explicit W_2 bounds ; regime-specific noise schedule $g_k(\tau)$.

— `FactorScenarioGenerator` : generation of N regime-conditional factor scenarios via GAN (stage 1) or SGM (stage 2) ; backtesting on 2008, 2020 and 2022 crises.

— `CentralScenarioSelector` : extraction of a representative central scenario (medoid / barycenter) from the generated scenario set for governance, budgeting and committee use.

4.2. Axis 2 — Supervised SCR proxy (Year 2)

Model. Let $\mathbf{P}_t \in \mathbb{R}^p$ be the liability state vector : effective duration, average guaranteed rate, observed lapse ratio, and the private asset vintage lag $\tilde{F}^{\mathrm{PA}}_{t-1}$ capturing NAV smoothing bias. We learn a supervised proxy :

$$\widehat{\mathrm{SCR}}(t) = f_\theta(\mathbf{F}_t,\, z_t,\, \mathbf{P}_t), \tag{4}$$

where f_θ is a neural network (MLP + attention) trained on a design-of-experiment $\{(\mathbf{F}^{(i)}, z^{(i)}, \mathbf{P}^{(i)}), \mathrm{SCR}^{(i)}\}$ computed offline via Prophet/MoSes. The key methodological innovation is **adaptive weighting** : scenarios close to the regulatory threshold and in stress regimes receive higher training weight, concentrating accuracy where the decision is most sensitive. The generalisation bound satisfies :

$$\left|\mathrm{SCR}(t) - f_\theta(\mathbf{F}_t, z_t, \mathbf{P}_t)\right| \;\le\; \varepsilon_k(n_k, \delta), \tag{5}$$

where $\varepsilon_k(n_k, \delta)$ decreases with the number of training points n_k in regime k and with confidence level $1 - \delta$.

Contribution. First SCR proxy jointly conditioned on the macroeconomic regime, TPA factors, and private asset NAV lag, with empirical error bounds validated across portfolio ranges including Fonds Euro and Euro-Croissance. Inference in < 5 ms versus several hours for the full Prophet/MoSes calculation. This proxy is not only a reporting object : it enters the downstream allocation programme as an explicit state-dependent capital constraint.

Python deliverable — `scr_proxy` module.

— `LiabilityDataPipeline` : Prophet/MoSes interface for automated actuarial design-of-experiment generation.

— `SCRProxyModel` : MLP + attention network trained on 5,000+ Generali scenarios with adaptive regime weighting.

— `SCRCalculator` : near-instantaneous SCR computation (< 5 ms) for Market, Interest Rate and Equity modules.

— Validation report : relative error and confidence intervals per regime ; stress tests on 2022 and novel scenarios.

4.3. Axis 3 — Joint allocation and deep hedging policy (Year 3)

Model. For each portfolio, a latent objective map $\psi_t = g_\eta(\mathbf{F}_t, z_t, \mathbf{P}_t, \widehat{\mathrm{SCR}}_t)$ identifies the dominant factor objective to optimise (for example duration control, inflation hedging, carry extraction, liquidity preservation, or capital efficiency) as a function of the current regime, liability state and capital position. This layer is what allows the framework to move from a static allocation rule to a genuinely dynamic allocation policy.

The agent state is $s_t = (\mathbf{F}_t, \hat{C}_{z_t}, z_t, \mathbf{P}_t, \widehat{\mathrm{SCR}}_t)$, where $\hat{C}_{z_t}$ is the regime-estimated copula from Axis 1 and $\widehat{\mathrm{SCR}}_t$ the proxy from Axis 2. The action is the pair $(\mathbf{w}_t, \mathbf{h}_t)$, where $\mathbf{w}_t \in \Delta^{d-1}$ is the TPA factor allocation vector and $\mathbf{h}_t \in \mathcal{H}$ is the hedging position (puts, overlays, convexity structures). The policy π maximises :

$$\max_\pi \; \mathbb{E}_\pi\left[\sum_{t=0}^{T}\Big(r_{\psi_t}(\mathbf{w}_t, \mathbf{h}_t, \mathbf{F}_{t+1}) - \lambda_1 c(\mathbf{h}_t) - \lambda_2\|\mathbf{w}_t - \mathbf{w}_{t-1}\|_1 - \lambda_3 \mathbf{1}[\Delta w_t^{\mathrm{PA}} > 0]\Big)\right], \tag{6}$$

subject to the explicit capital constraint

$$\widehat{\mathrm{SCR}}_t \le \bar{C},$$

where r_{ψ_t} is the portfolio-specific objective functional selected in factor space, $\bar{C}$ the regulatory capital budget, $c(\mathbf{h}_t)$ the hedging cost (option premium, bid-ask spread), and the term $\lambda_3 \mathbf{1}[\Delta w_t^{\mathrm{PA}} > 0]$ penalises new private asset commitments to reflect their irrevocable, illiquid nature.

Integrated deep hedging. Unlike classical deep hedging (Buehler et al., 2019) which targets payoff replication, the hedging component here aims to reduce the $\mathrm{CVaR}_{0.99}$ of the capital deficit while internalising hedging costs. The regime-conditional factor dependence of Axis 1 is directly embedded in the simulation environment, forcing the policy to be robust to dependence regime switches — including the private asset delayed correction dynamic.

Python deliverable — `dl_allocator` module.

— `TPA_Environment` : Gym environment coupling `factor_engine` and `scr_proxy` ; simulates regime-switching trajectories including private asset NAV dynamics.

— `AllocationHedgingAgent` : PPO/SAC agent with dual head (factor allocation + hedging) and dynamic objective selection head ; trained on 10+ years of simulated multi-regime data.

— `RebalancingTrigger` : automated detection of trigger events (regime change, SCR threshold breach, target drift $> \Delta$) ; `GenAMBridge` interface for automated order execution.

— `StrategyBacktester` : rolling and expanding-window out-of-sample backtests on historical and simulated allocations, including 2008, 2020, 2022 ; RoRC, regime-conditional Sharpe, SCR CVaR, turnover and hedging cost metrics.

5. SYSTEM ARCHITECTURE

All modules are packaged in the Python library **`solvency-tpa`**, deployed in production at Generali with Sphinx documentation and pytest unit tests.

Module 1 `factor_engine`	**Module 2** `scr_proxy`	**Module 3** `dl_allocator`
HMMRegimeFilter (SMC) RegimeConditionedGAN RegimeConditionedSGM FactorScenarioGen. / CentralScenarioSel.	LiabilityDataPipeline SCRProxyModel (MLP + attention) SCRCalculator	TPA_Environment (Gym) AllocationHedging Agent (PPO/SAC) RebalancingTrigger / StrategyBacktester
Input : Bloomberg rates, spreads, vol, private NAV lags *Output :* regime-conditional scenarios + central scenario + + copulas $\hat{C}_k$	*Input :* Prophet/ MoSes + M1 factors + private NAV lag *Output :* SCR in < 5 ms + error bounds per regime + optimisation constraint	*Input :* M1 + M2 outputs *Output :* allocation + hedge + GenAM orders + OOS backtests

Tech stack : Python 3.11+, PyTorch (neural proxy and RL agent), pyvinecopulib / scipy (copulas), `particles` (SMC, Chopin), `gymnasium` / `stable-baselines3` (RL), Bloomberg API. GPU cluster for training ; CPU for production inference.

6. EXPECTED CONTRIBUTIONS

Type	**Description**
Theoretical	(i) Non-asymptotic error bounds on estimated conditional tail dependence λ_k for regime-conditional copulas and GANs (Fermanian) ; (ii) Extension of W_2 bounds of Gentiloni-Silveri & Ocello (ICML 2025) to the conditional SGM case with online SMC regime estimation (Ocello) ; (iii) Generalisation bounds for the regime-conditioned SCR proxy with adaptive weighting and private asset NAV lag.
Methodological	(i) HMM-SMC architecture coupled to a regime-conditional GAN/SGM with private asset vintage lag factor and central scenario extraction (module 1) ; (ii) Supervised SCR proxy conditioned on regime and TPA factors with adaptive weighting and explicit integration as optimisation constraint (module 2) ; (iii) Dual allocation-and-hedging agent under private asset engagement constraints, automatic portfolio-objective identification and out-of-sample backtesting (module 3).
Industrial	Python library `solvency-tpa` in production at Generali : auditable regime-conditional scenario generation with a central baseline scenario, SCR in < 5 ms as optimisation constraint, TPA allocation with integrated hedging, OOS backtesting and GenAM interface.
Regulatory	Models with controlled, auditable guarantees — W_2 bounds for the generative module, empirical error bounds for the SCR proxy — directly usable by the ACPR and the Generali Group Risk Division.

7. TIMELINE (36 MONTHS)

Phase	**Period**	**Key deliverables**
1	1st year	Bloomberg factor database (20 years) including private asset NAV lags ; online HMM-SMC calibration in production ; regime-conditional GAN with copula validation on historical crises ; representative central scenario extraction ; `factor_engine` v1.0 ; paper submission (Journal of Multivariate Analysis or Finance & Stochastics).
2	2nd year	Actuarial design-of-experiment (Prophet/MoSes) ; training and validation of regime-conditioned SCR proxy with private asset NAV lag ; integration of the SCR proxy as an explicit optimisation constraint ; empirical error bounds per regime and portfolio range ; `scr_proxy` v1.0 ; paper submission (Insurance : Mathematics and Economics).
3	3rd year	Training of dual allocation-and-hedging agent with private asset engagement constraints and automatic objective selection per portfolio ; full three-module coupling ; multi-regime out-of-sample backtesting (2008, 2020, 2022) ; `dl_allocator` v1.0 ; GenAM interface in production ; PhD defence and publications (Quantitative Finance or JFQA).

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